

Machine Learning (CE 40477)
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- ① Unsupervised Learning Overview
- ② K-Means
- ③ Challenges in K-Means
- ④ Other Clustering Algorithms

1 Unsupervised Learning Overview

2 K-Means

3 Challenges in K-Means

4 Other Clustering Algorithms

Unsupervised Learning

- **Unsupervised Learning** involves analyzing unlabeled data to uncover hidden patterns or structures within the data

Some Common Tasks

- **Clustering:** Grouping data points into clusters based on similarity.
- **Dimensionality Reduction:** Reducing the number of features under consideration and keeping (perhaps approximately) the most informative features.
- **Anomaly Detection:** Identifying data points that deviate significantly from the norm (e.g., fraud detection).
- **Generative Modeling:** Learning the distribution of data to generate new, similar instances.

Clustering

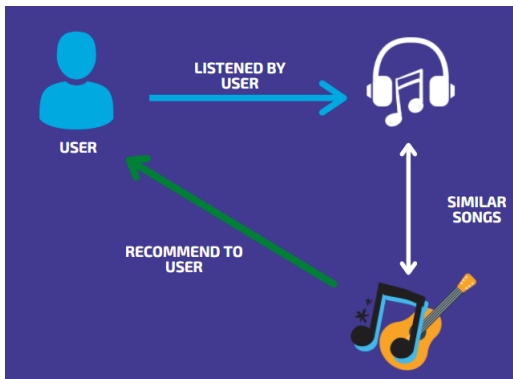
- Clustering organizes data points into groups of similar objects.
- Data points in a cluster are more similar to each other than to those in other clusters.
- The notion of similarity depends on the task at hand (e.g., purchase behavior in market segmentation).

Some Applications of Clustering

- Customer Segmentation (Marketing)
- Image Segmentation and Object Detection (Computer Vision)
- Anomaly Detection (Cybersecurity, Finance)
- Genomics and Bioinformatics
- Social Network Analysis and Community Detection

Clustering in Action: Music Recommendation Systems

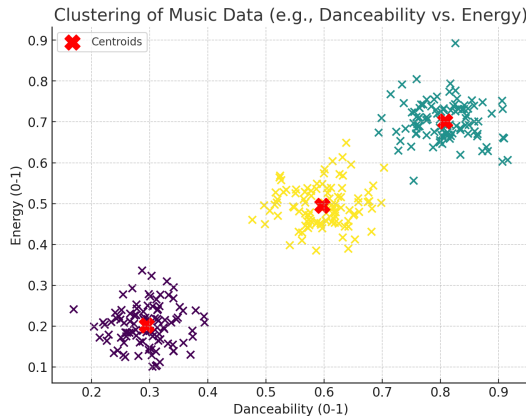
- Music recommendation systems cluster songs based on similarity.



Adopted from machinelearninggeek.com

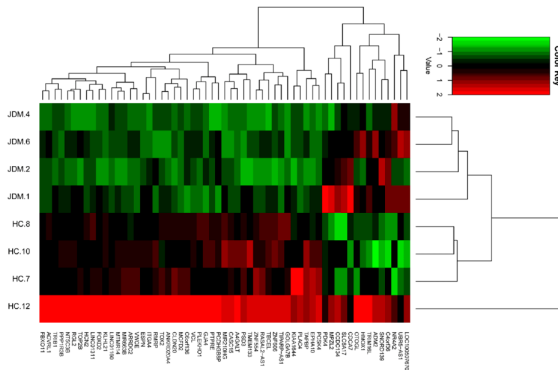
Clustering in Action: Music Recommendation Systems

- When you like a song, the system suggests others from the same cluster.



Clustering in Action: Gene Expression Clustering

- Clustering can decipher hidden patterns in gene expression data, which can help in understanding disease mechanisms or genetic variations.



Two Beginning Questions

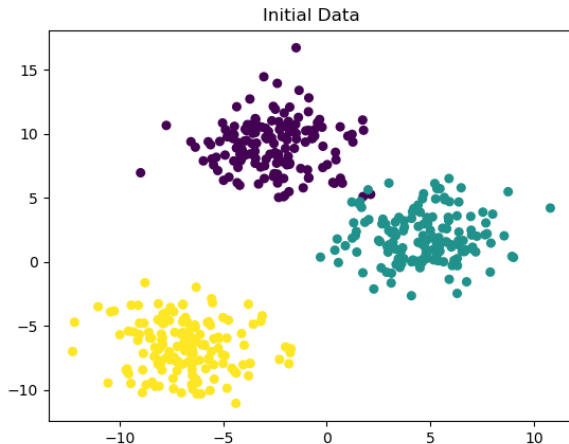
- How to create 'good' clusters?
- How many clusters do we need?

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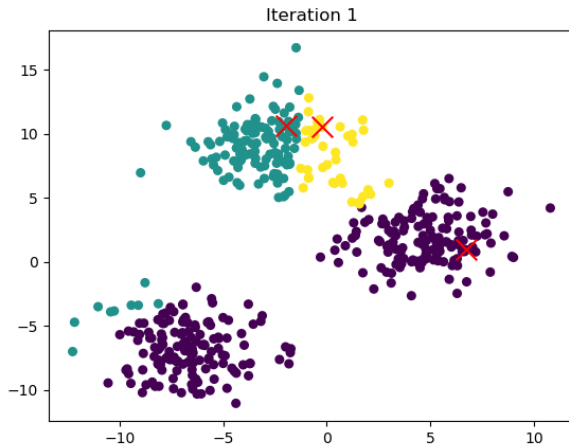
K-Means overview

- The most widely used clustering algorithm.
- Partitions data into K distinct groups based on feature similarity
- It works by **iteratively** assigning data points to the nearest centroid (mean of the group) and then recalculating the centroids based on the new group memberships
- The process repeats until the assignments no longer change

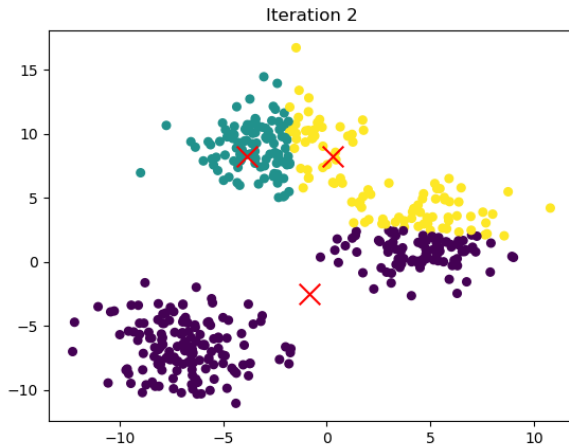
K-Means in action



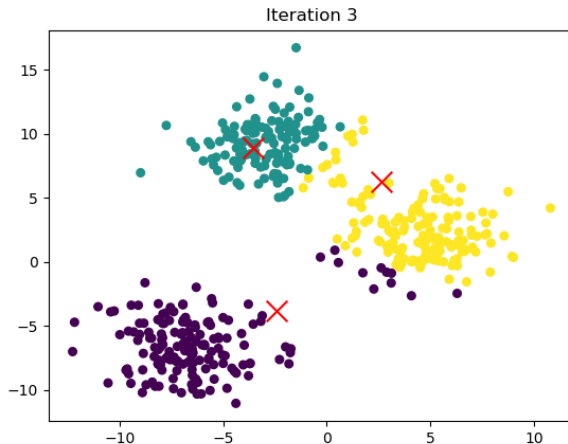
K-Means in action



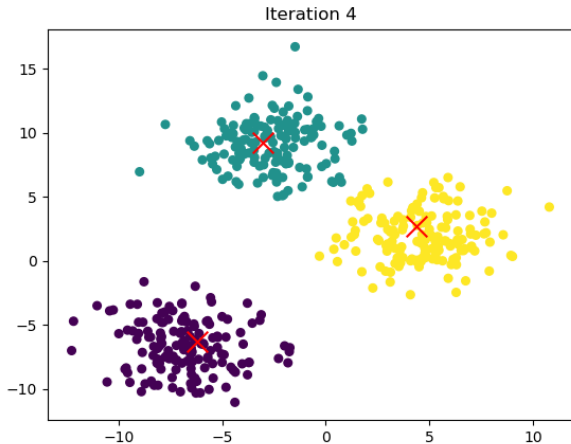
K-Means in action (cont.)



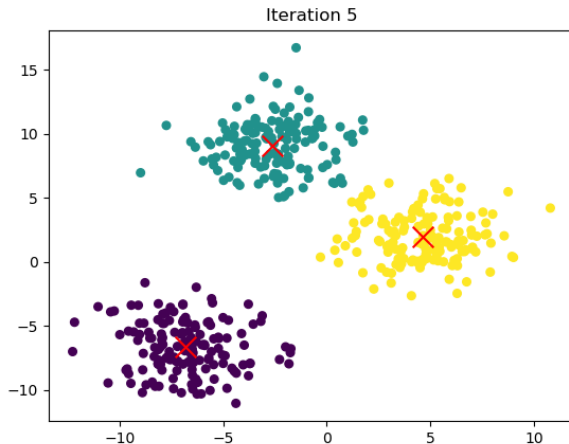
K-Means in action (cont.)



K-Means in action (cont.)



K-Means in action (cont.)



Algorithm

Algorithm 1 K-means Clustering

- 1: **Input:** K (number of clusters), $D = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}\}$ (data points)
 - 2: **Initialize:** Select K random points as centroids $\{\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K\}$
 - 3: **repeat**
 - 4: Assign each point $\mathbf{x}^{(i)}$ to nearest centroid $f(\mathbf{x}^{(i)}) = \arg \min_j \|\mathbf{x}^{(i)} - \boldsymbol{\mu}_j\|$
 - 5: For each $1 \leq j \leq K$ set $C_j = \{\mathbf{x}^{(i)} | f(\mathbf{x}^{(i)}) = j\}$
 - 6: Update centroids $\boldsymbol{\mu}_j = \frac{1}{|C_j|} \sum_{\mathbf{x}^{(i)} \in C_j} \mathbf{x}^{(i)}$
 - 7: **until** Centroids do not change
 - 8: **Output:** Final clusters $\{C_1, C_2, \dots, C_K\}$
-

Problem definition

- Formally: We have $X_{\text{train}} = \{x^{(1)}, x^{(2)}, \dots, x^{(N)}\} \subseteq \mathbb{R}^d$
- K is the number of clusters.
- We are learning:
 - ① A function or mapping $f: \mathbb{R}^d \rightarrow \{1, 2, \dots, K\}$ that assigns a cluster to each data point.
 - ② A set of K prototypes $\mu = \{\mu_1, \mu_2, \dots, \mu_K\} \subseteq \mathbb{R}^d$ as the cluster representatives, called **centroids**.

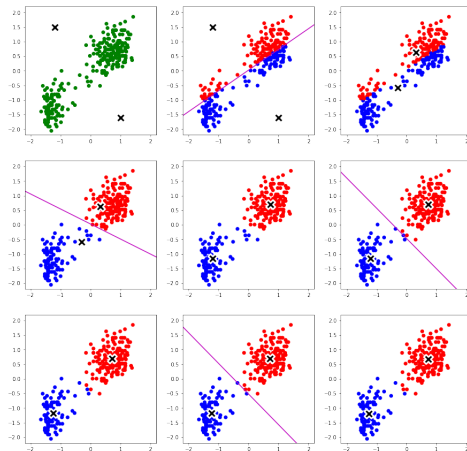
Objective Function

- We want samples in the same cluster to be similar.
- In K-Means, this is expressed as:

$$J = \sum_{j=1}^K \sum_{x^{(i)} \in C_j} \|x^{(i)} - \mu_j\|^2$$

- Choose f and $\mu = \{\mu_1, \mu_2, \dots, \mu_K\}$ to minimize this.
- This problem is NP-hard. K-Means is a heuristic solution, which is NOT guaranteed to find optimal solution.

K-Means Process Example



Adopted from [mlbhanuyerra.github.io](https://github.com/mlbhanuyerra)

Convergence

- How do we know K-Means will converge in a finite number of steps ?
- First we show in each step J will decrease, as long as we have not converged.

Convergence (cont.)

- We initially assign each sample to the nearest centroid.

$$f(x) := \operatorname{argmin}_j \|x - \mu_j\|^2$$

.

- Keep each sample's assignment fixed until a closer centroid is found.
- Each time a sample is reassigned, the total distance between samples and their centroids decreases.
- The number of possible sample-to-centroid assignments is finite.
- The algorithm terminates when no sample changes its assigned centroid.

Convergence (cont.)

- In Updating step, with $f(x)$ fixed, J is a quadratic function of μ_j (like SSE) and by taking derivative we can minimize it as:

$$\frac{\partial J}{\partial \mu_j} = 0 \implies \sum_{x^{(i)} \in C_j} 2(x^{(i)} - \mu_j) = 0$$

- This means we should **update** each μ_j as the mean of cluster C_j :

$$\mu_j = \frac{\sum_{x^{(i)} \in C_j} x^{(i)}}{|C_j|}$$

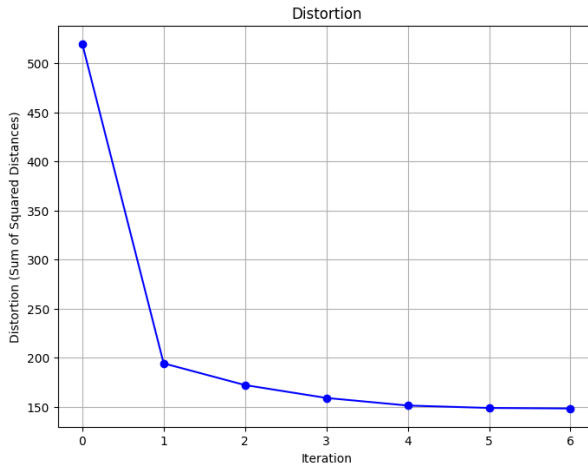
Convergence (cont.)

- For each cluster, the mean of its samples minimizes squared distances.
- For C_j if μ'_j was the old centroid we have: $\sum_{x^{(i)} \in C_j} \|x^{(i)} - \mu'_j\|^2 \geq \sum_{x^{(i)} \in C_j} \|x^{(i)} - \mu_j\|^2$. So $J_{\text{new}} \leq J_{\text{old}}$.

Convergence (cont.)

- J is non-negative, and there are a finite number of partitions so there is a minimum for J and we can't decrease J forever.
- Therefore we must converge at some point.
- The convergence properties of the K-means algorithm were studied by MacQueen (1967).

K-Means Convergence (cont.)



Strengths

- Simple: easy to understand and to implement.
- Efficient: Time complexity: $O(tkn)$, where
 - n is the number of data points,
 - k is the number of clusters, and
 - t is the number of iterations.

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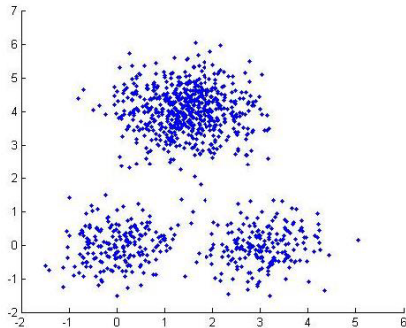
Initialization

- K-Means always converges. What could go wrong ?
- K-Means algorithm is a **heuristic**
- It requires initial centroids, and the choice is important as it could affect the t in $O(tkn)$.

Local Optimum

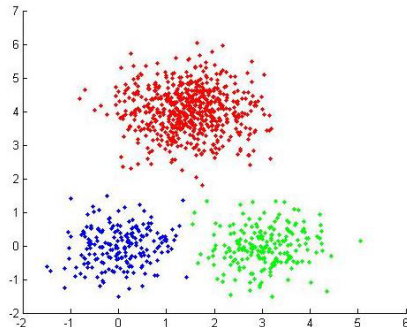
- The algorithm finds a local minimum but there is no guarantee to find global minimum.
- Its result is highly affected by the initialization.
- Some suggestions are:
 - Multiple runs with random initial centroids, then select the "best" result.
 - Initialization heuristics (K-Means++ , Furthest Traversal).
 - Initializing with the suggested results of another method.

Local Optimum

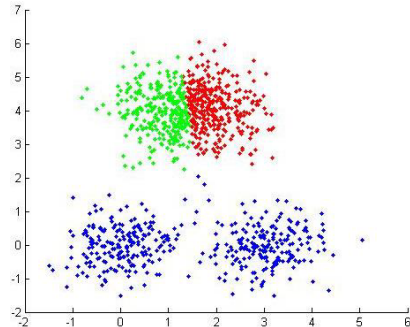


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Local optimum (cont.)



Optimal clustering

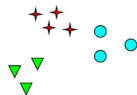


Possible clustering

Definition of Mean

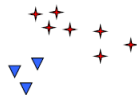
- We assume $x^{(i)} \in \mathbb{R}^d$, which is not always the case. K-Means requires a space where sample **mean** is defined.
 - Categorical data.
 - A suggested solution: K-Mode - the centroid is the most frequent category (the mode) in each cluster.
 - Closest centroid is found by the Hamming Distance.

How many clusters?



How many clusters?

Six Clusters



Two Clusters

Four Clusters

Adopted from

slides of Dr. Soleymani, Modern Information Retrieval Course, Sharif University of technology.

How many clusters? (cont.)

- Number of clusters is usually given in advance in the problem of clustering. However; finding the right number of clusters is also a problem.
- First we need to know how we can evaluate a clustering.

Summary of Methods

Method	Objective	Interpretation for Optimal K
Inertia (Elbow Method)	Minimize the sum of squared distances (inertia).	"Elbow" point where the curve flattens.
Silhouette Score	Maximize the mean silhouette score.	Maximize the score close to 1.
Davies-Bouldin Index	Minimize the Davies-Bouldin index.	Minimize the DB index.

Each method offers a different perspective on determining the optimal number of clusters, and it is often useful to use a combination of methods to get a more comprehensive understanding of the cluster structure in the data.

Clustering Evaluation

- Evaluating clusters involves two key aspects:
 - **Intra-cluster cohesion (compactness):** How similar the data points are within a cluster.
 - Often measured by the within-cluster sum of squares (WCSS):

$$WCSS = \sum_{i=1}^K \sum_{x \in C_i} ||x - \mu_i||^2$$

Clustering Evaluation

- **Inter-cluster separation (isolation):** How different the data points are between clusters.
 - Single-link (Minimum Distance):
 - Measures the ****minimum distance**** between any two points from different clusters.

$$d_{\text{single}}(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$

- Complete-link (Maximum Distance):
- Measures the maximum distance between any two points from different clusters.

$$d_{\text{complete}}(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$

Clustering Evaluation

- **Inter-cluster separation (isolation):** How different the data points are between clusters.
 - Centroid (Wards Method):
 - Measures the distance between the centroids of two clusters.

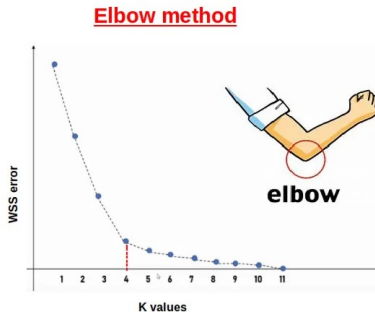
$$d_{\text{centroid}}(C_i, C_j) = d(\mu_i, \mu_j)$$

- Average-link:
- Measures the average distance between all pairs of points from different clusters.

$$d_{\text{average}}(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d(x, y)$$

Elbow Method for Optimal K

- Finds the optimal number of clusters K by minimizing the within-cluster sum of squares (WCSS).
- Elbow Point:
 - Plot WCSS versus K .
 - The point where the rate of decrease sharply slows down (resembles an "elbow") is considered the optimal K .



Silhouette Method for Cluster Evaluation

- Silhouette Score for a single point i :

$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

- where:
 - $a(i)$ is the average distance between i and all other points in the same cluster.
 - $b(i)$ is the average distance between i and points in the nearest neighboring cluster.
- Interpretation:
 - $S(i) \in [-1, 1]$
 - $S(i) \approx 1$: Well-clustered.
 - $S(i) \approx 0$: On or near the decision boundary between clusters.
 - $S(i) \approx -1$: Misclustered.

How many Clusters? (cont.)

- There is a trade-off between having better focus within each cluster or having too many clusters.
- Don't want one-element clusters.
- **Optimization problem:** penalize having too many clusters

$$K^* = \arg \min_k J(k) + \lambda k$$

Outliers

- The algorithm is sensitive to outliers
- Outliers are data points that are very far away from other data points.
- Outliers could be errors in data recording or unique data points with significantly different values.

Data Distribution

- There is a problems with how k-means defines clusters.
- K-means assumes clusters are spherical and separated by equal variance, which limits its effectiveness on non-spherical or complex-shaped clusters.

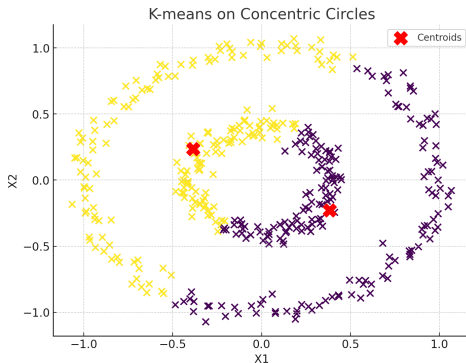


Figure 1: example when k-means wont work

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Hard vs Soft Clustering

- **Hard Clustering(Partitional):** Each data point belongs to exactly one cluster
 - More common and easier to use.
- **Soft Clustering(Bayesian)**

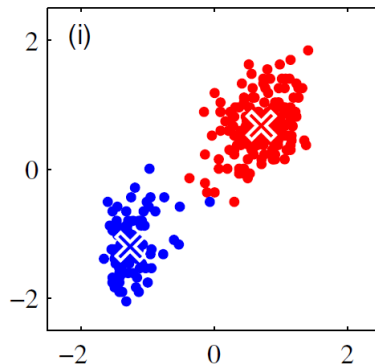


Figure adapted from Machine Learning and
Pattern Recognition, Bishop

Hard vs Soft Clustering (cont.)

- **Hard Clustering(Partitional)**
- **Soft Clustering(Bayesian):** Each sample is assigned to different clusters with probabilities, rather than $\{0, 1\}$.
 - data point belongs to each cluster with a probability

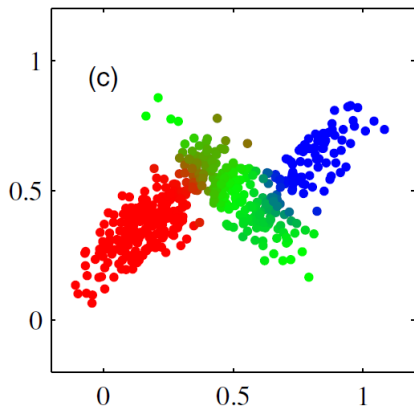
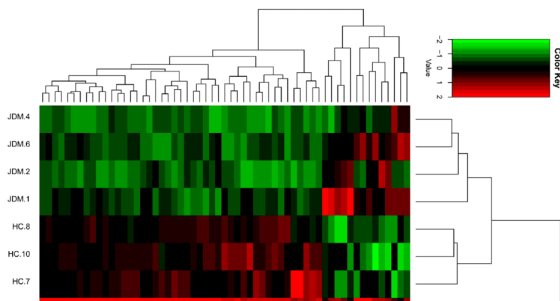


Figure adapted from Machine Learning and
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Hierarchical Clustering

- **Hierarchical** algorithms find successive clusters using previously established clusters. Two Types:
 - **Agglomerative (bottom-up)**: Start with individual points and merge clusters.
 - **Divisive (top-down)**: Start with all points and split clusters.

Result: A hierarchy of clusters represented by a dendrogram.

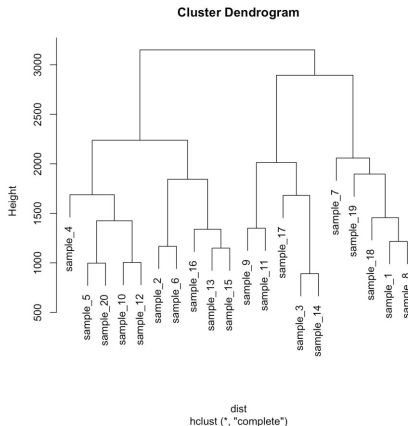


Agglomerative Clustering Algorithm

- Start with each point as its own cluster.
- Merge the "closest" clusters.
- Repeat until one cluster remains or desired number is reached.
- Closest cluster can be determined using inter-cluster separation measures

Dendrogram and Cutting

- A dendrogram shows the hierarchy of merges.
- Cut the dendrogram at a desired level to form clusters.



Adopted from r-graph-gallery.com

Hierarchical Algorithms

- Advantages:
 - No need to specify the number of clusters.
 - Produces a dendrogram for visualization.
 - Works with arbitrary-shaped clusters.
- Disadvantages
 - High computational cost.
 - Sensitive to noise and outliers.
 - Greedy: cannot undo merges.

DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise):

- Groups points in high-density regions.
- Labels points in low-density regions as noise.
- Does not require specifying the number of clusters K .

Parameters:

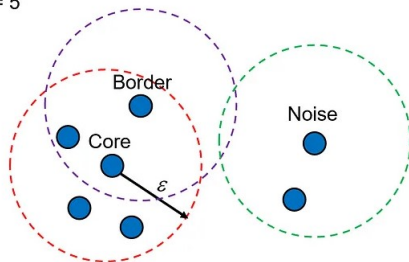
- ϵ (epsilon): Maximum distance for neighbors.
- minPts: Minimum points to form a dense region.

Core Concepts in DBSCAN

DBSCAN defines three types of points:

- **Core Point:** A point with at least minPts neighbors within distance ϵ .
- **Border Point:** A point within ϵ of a core point but with fewer than minPts neighbors.
- **Noise:** Points that are neither core points nor border points.

MinPts = 5



Adopted from ai.plainenglish.io

Core Concepts in DBSCAN (cont.)

Definitions:

- A point x_i is a core point if:

$$|\{x_j : d(x_i, x_j) \leq \epsilon\}| \geq \text{minPts}$$

- A point is a border point if it is within distance ϵ of a core point, but not itself a core point.

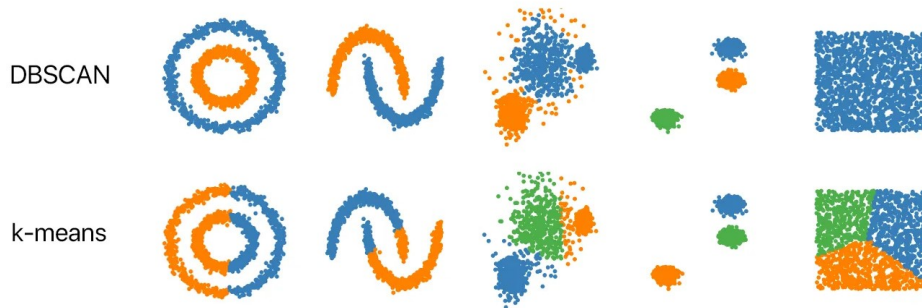
DBSCAN Algorithm Steps

Algorithm Steps:

- ① For each unvisited point x_i :
 - Mark x_i as visited.
 - Find all points within distance ϵ (neighborhood).
- ② If x_i is a core point:
 - Create a new cluster and expand it by recursively adding all reachable core and border points.
- ③ If x_i is not a core point:
 - Label it as noise if it does not belong to any cluster.

Advantages of DBSCAN

- Can find clusters of arbitrary shape (non-spherical).
- Does not require specifying the number of clusters K in advance.
- Robust to noise and outliers.
- Works well with large datasets.



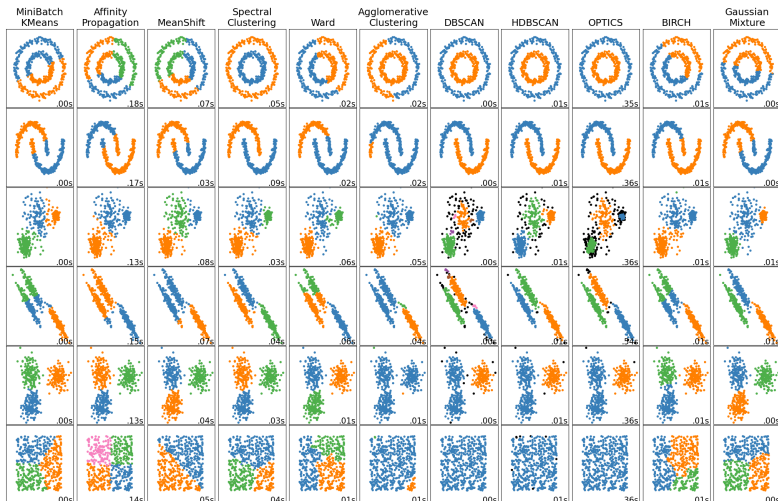
Adopted

Limitations of DBSCAN

- DBSCAN struggles with datasets of varying densities.
- Sensitive to the selection of parameters ϵ and minPts .
- Does not perform well with high-dimensional data.

Clustering Algorithms

- Each algorithm is suited for different kinds of patterns and information in data.



Contributions

- **This slide has been prepared thanks to:**
 - Hooman Zolfaghari

